MAT 320 Final Project - Marcus's Car

Hannah Pratsch & Marcus Larsen

2026-05-04

Abstract

This project analyzes a long term observational data set by tracking fuel usage and milage for a 2014 Subaru Forester from 2014 to 2026. The data, recorded at each fuel fill-up, allow for an examination of trends in gas prices, fuel efficiency, and driving behavior over time. Seasonal patterns in both dollars per gallon and miles per gallon are analyzed using one-way ANOVA tests, revealing statistically significant differences across seasons. Changes in driving behavior are explored by comparing miles driven per day across different ownership periods: under Marcus's grandparents, Marcus's ownership before Amber, and Marcus's ownership after Amber. Because the distribution of daily milage is skewed, Wilcoxon Rand Sum tests are used to assess the differences between these ownership groups. The results indicate that both fuel efficiency and gas prices vary significantly by season, and that driving distance increased substantially during later ownership periods, particularly after changes in work and travel patterns. Overall, this study demonstrates how long term, real world data can reflect both environmental conditions and personal lifestyle changes.

Introduction

This project looks at a long-term, real-life dataset that tracks the mileage and fuel fill-ups of a 2014 Subaru Forester. The records begin in 2014 when Marcus's grandparents bought the car and mainly used it for city driving, carefully writing down each fill-up in a small journal. In 2022, Marcus took over the vehicle and continued the journal, first using his vehicle for regular trips to Lakeland University and later, starting in 2025, for more frequent and longer drives related to his job (including travel to places like Brown Deer and Sturgeon Bay) and visits to his girlfriend in Manitowoc. Because the data were simply recorded as the car was used, this is an observational data set rather than an experiment, but it is consistent and detailed due to the careful record-keeping. Using this information, the goal is to examine how gas prices (dollars per gallon), fuel efficiency (miles per gallon), and overall driving patterns have changed over time, especially across these different phases of use, and to compare Marcus's driving habits with those of his grandparents.

The main research questions of this data focus on identifying trends and changes over time in:
- Cost of gasoline over time (dollars per gallon)
- Fuel efficiency over time (miles per gallon)
- Driving frequency and distance (miles driven across different time periods)

Method and Results

The analysis uses the data collected from the records kept in the journal from Marcus and his grandparents tracking the date, mileage, gallons, and cost of each car fill-up. Using this data, we have also calculated miles per gallon, dollars per gallon, ownership, and season of each data entry.

The libraries and data set we will be working with:

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.4.3

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
carData <- read.csv("MAT 320 Data.csv")
```

To get a basic understanding of the data we look at the following graphs and calculations.

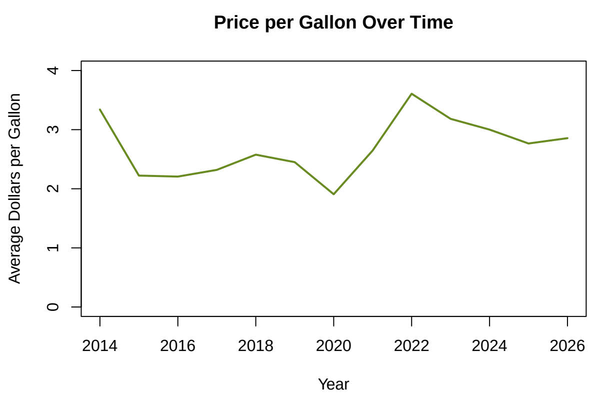
```
entries <- nrow(carData)
totalDollars <- sum(carData$Cost, na.rm = TRUE)
totalMiles <- 156135
dollarsPerMile <- totalDollars / totalMiles
avgDollars <- sprintf("%.2f", mean(carData$DPG, na.rm = TRUE))
avgMileage <- sprintf("%.2f", mean(carData$MPG, na.rm = TRUE))

dollarsPerMile <- round(dollarsPerMile, 2)
totalDollars <- sprintf("%.2f", totalDollars)
```

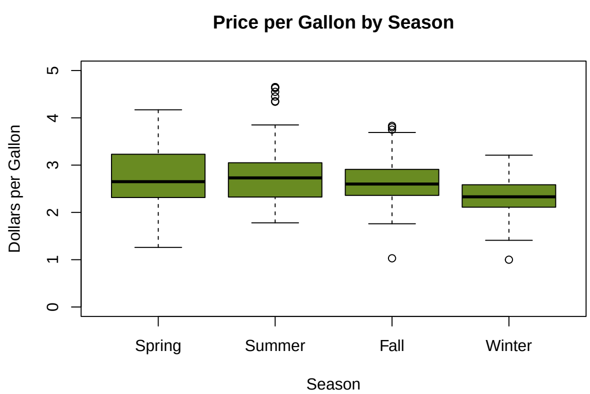
Our data set spans from March 7th 2014 to April 25th 2026. Below are some interesting summary statistics:
- The total number of entries for our data set is 669.
- The total dollars spent on gas is $15645.28.
- The total miles on the car is 156,135.
- This means that on average each mile cost approximately $0.1.
- The total average dollars per gallon is $2.63.
- The total average miles per gallon is 26.67.

First we will show the average Dollars per Gallon per year and average Dollars per Gallon per season.

Note that the seasons have been created as follows:
0 = Spring (March, April, May)
1 = Summer (June, July, August)
2 = Fall (September, October, November)
3 = Winter (December, January, February)

```
AvgDPGPerYear <- aggregate(DPG ~ Year, data = carData, FUN = mean)

plot(AvgDPGPerYear$Year, AvgDPGPerYear$DPG,
     type = "l", col = "olivedrab4", lwd = 2,
     main = "Price per Gallon Over Time",
     xlab = "Year", ylab = "Average Dollars per Gallon",
     ylim = c(0, 4))

boxplot(DPG ~ Season, data = carData,
        col = "olivedrab4",
        main = "Price per Gallon by Season",
        xlab = "Season", ylab = "Dollars per Gallon",
        ylim = c(0, 5),
        names = c("Spring", "Summer", "Fall", "Winter"))
```

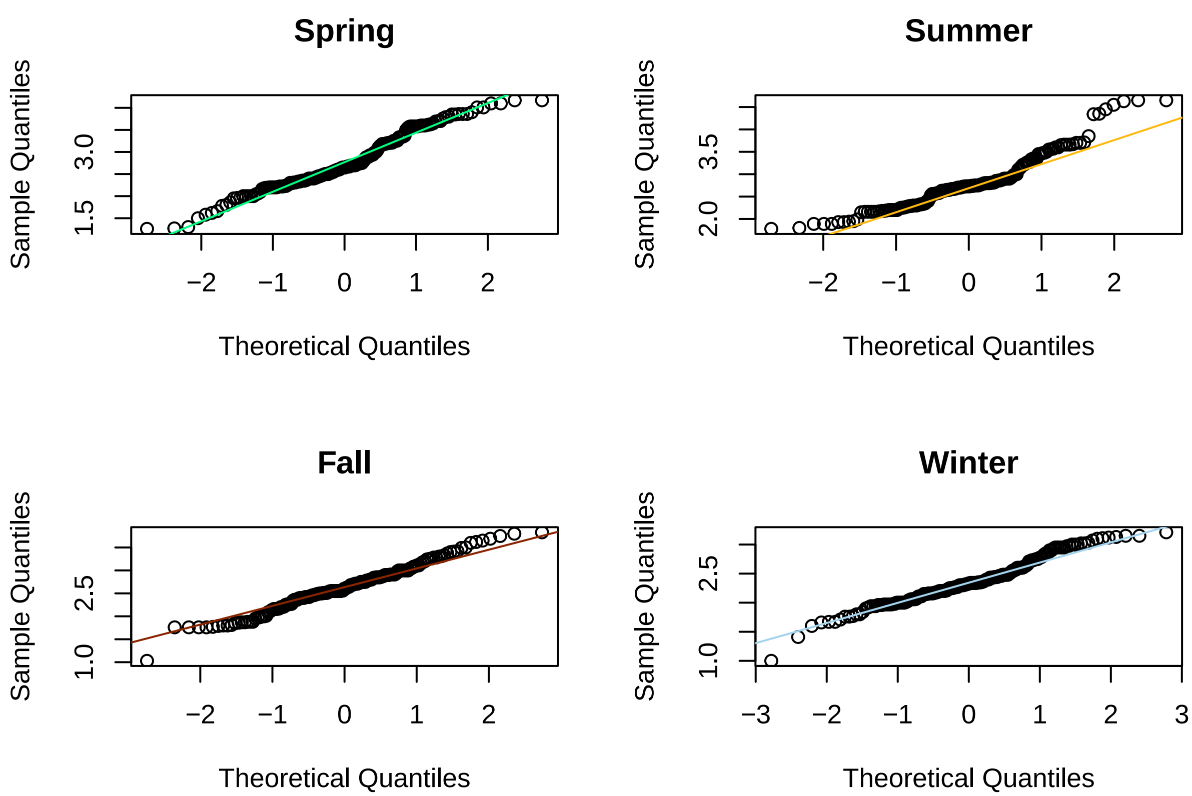




From these graphs we can see there may be a significant difference in gas prices between seasons. Lets find out by using a one-way ANOVA. First we need to check our assumptions.

```
spring <- carData$DPG[carData$Season == 0]
summer <- carData$DPG[carData$Season == 1]
fall   <- carData$DPG[carData$Season == 2]
winter <- carData$DPG[carData$Season == 3]

par(mfrow = c(2, 2))

qqnorm(spring, main = "Spring")
qqline(spring, col = "springgreen")

qqnorm(summer, main = "Summer")
qqline(summer, col = "darkgoldenrod1")

qqnorm(fall, main = "Fall")
qqline(fall, col = "orangered4")

qqnorm(winter, main = "Winter")
```

```r
qqline(winter, col = "lightskyblue2")
```




```r
sd_vals <- c(sd(spring), sd(summer), sd(fall), sd(winter))
max(sd_vals) / min(sd_vals)
```

```
## [1] 1.699685
```

All assumptions are met since we have an independent simple random sample, the population is normally distributed as shown by the Q-Q plots, and we have similar standard deviations since the ratio of the largest to the smallest sample standard deviation is less than 2. It is appropriate to apply one-way ANOVA procedures to this data using the following hypotheses:

H0: The mean dollars per gallon is the same for all four seasons Ha: At least one season has a different mean dollars per gallon

```r
dollarsAnova <- aov(DPG ~ factor(Season), data = carData)
summary(dollarsAnova)
```

```
##                 Df Sum Sq Mean Sq F value   Pr(>F)    
## factor(Season)   3  21.75    7.25   24.98 2.52e-15 ***
## Residuals      665 193.00    0.29                     
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

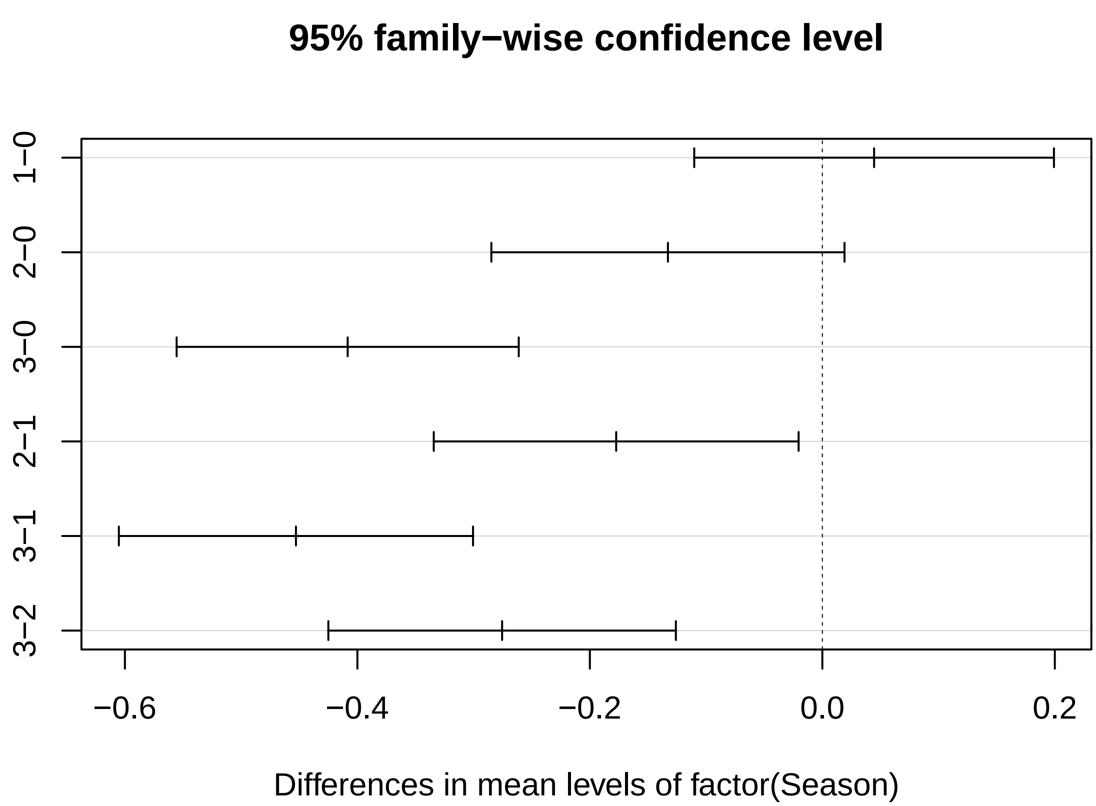
At the 5% significance level there is enough evidence to reject the null hypothesis and conclude that a difference exists among the mean price per gallon of gas during different seasons. The p-value is a lot less than 0.05 so we reject the null hypothesis of them being the same.

Next lets see which seasons are different.

```r
TukeyHSD(dollarsAnova)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = DPG ~ factor(Season), data = carData)
##
## $`factor(Season)`
##            diff        lwr         upr     p adj
## 1-0  0.04451525 -0.1102169  0.19924742 0.8804933
## 2-0 -0.13284597 -0.2847550  0.01906305 0.1105064
## 3-0 -0.40835187 -0.5555099 -0.26119385 0.0000000
## 2-1 -0.17736121 -0.3343101 -0.02041232 0.0194755
## 3-1 -0.45286712 -0.6052222 -0.30051199 0.0000000
## 3-2 -0.27550590 -0.4249930 -0.12601881 0.0000150
```

```r
plot(TukeyHSD(dollarsAnova))
```

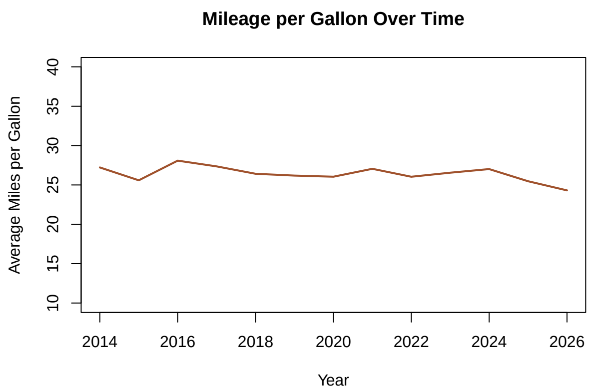


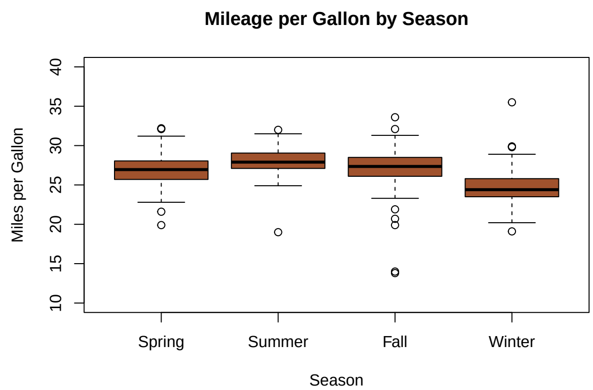
There is enough evidence to show a difference in gas prices between fall and winter, summer and winter, summer and fall, and spring and winter. There is not enough evidence to show a difference in gas prices between spring and fall or spring and summer since these confidence intervals contain zero. However it is worth noting that the spring and fall interval just bleary contains zero.

Next we will show the average Mileage per Gallon per year and average Mileage per Gallon per season.

```r
AvgMPGPerYear <- aggregate(MPG ~ Year, data = carData, FUN = mean)

plot(AvgMPGPerYear$Year, AvgMPGPerYear$MPG,
```

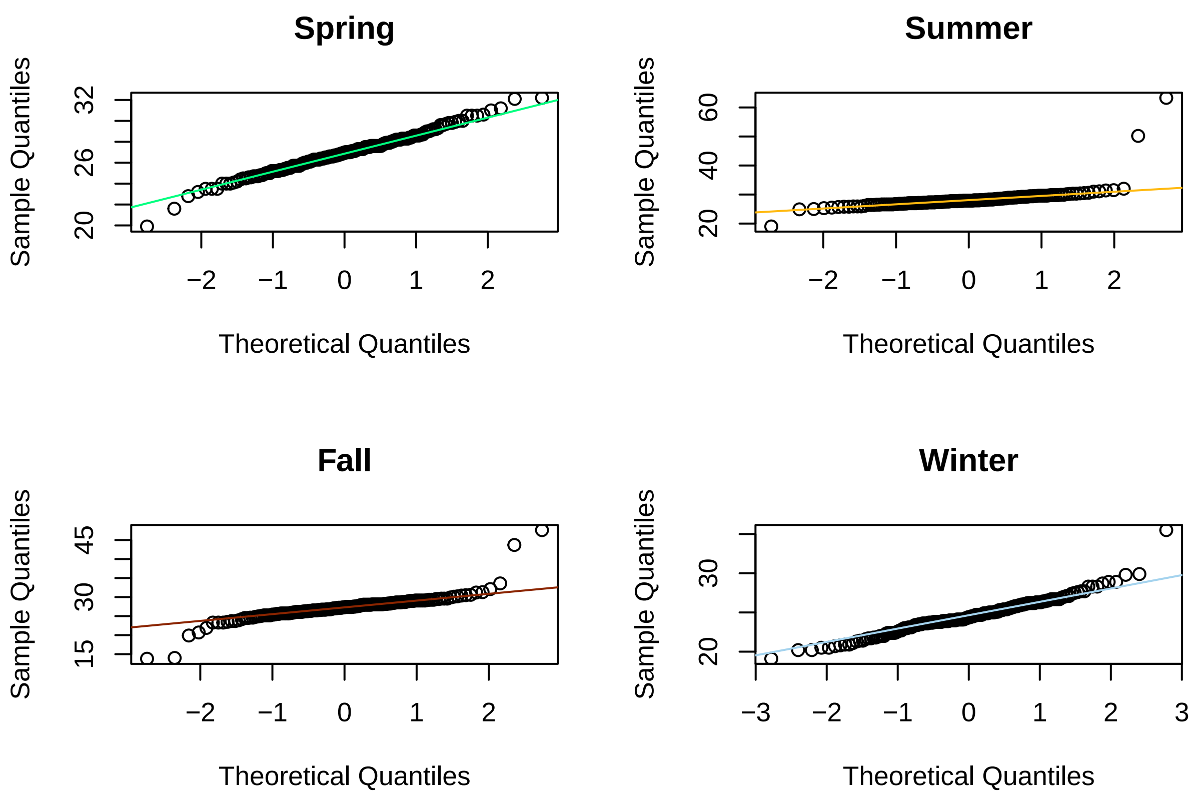
```r
      type = "l", col = "sienna", lwd = 2,
      main = "Mileage per Gallon Over Time",
      xlab = "Year", ylab = "Average Miles per Gallon",
      ylim = c(10,40))

boxplot(MPG ~ Season, data = carData,
        col = "sienna",
        main = "Mileage per Gallon by Season",
        xlab = "Season", ylab = "Miles per Gallon",
        ylim = c(10, 40),
        names = c("Spring", "Summer", "Fall", "Winter"))
```





From these graphs we can see there may be a significant difference in gas mileage between seasons. Lets find out by using a one-way ANOVA. First we need to check our assumptions.

```r
spring <- carData$MPG[carData$Season == 0]
summer <- carData$MPG[carData$Season == 1]
fall   <- carData$MPG[carData$Season == 2]
winter <- carData$MPG[carData$Season == 3]

par(mfrow = c(2, 2))

qqnorm(spring, main = "Spring")
qqline(spring, col = "springgreen")

qqnorm(summer, main = "Summer")
qqline(summer, col = "darkgoldenrod1")

qqnorm(fall, main = "Fall")
qqline(fall, col = "orangered4")

qqnorm(winter, main = "Winter")
qqline(winter, col = "lightskyblue2")
```




```r
sd_vals <- c(sd(spring), sd(summer), sd(fall), sd(winter))
max(sd_vals) / min(sd_vals)
```

```
## [1] 1.969179
```

All assumptions are met since we have an independent simple random sample, the population is normally distributed as shown by the Q-Q plots, and we have similar standard deviations since the ratio of the largest to the smallest sample standard deviation is less than 2. It is appropriate to apply one-way ANOVA procedures to this data. I would like to point out that the ratio is close to 2 being at 1.96158. We will use the following hypotheses:

H0: The mean miles per gallon is the same for all four seasons Ha: At least one season has a different mean miles per gallon

```r
milesAnova <- aov(MPG ~ factor(Season), data = carData)
summary(milesAnova)
```

```
##                  Df Sum Sq Mean Sq F value Pr(>F)
## factor(Season)   3   1335   445.1   57.41 <2e-16 ***
## Residuals      665   5156     7.8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

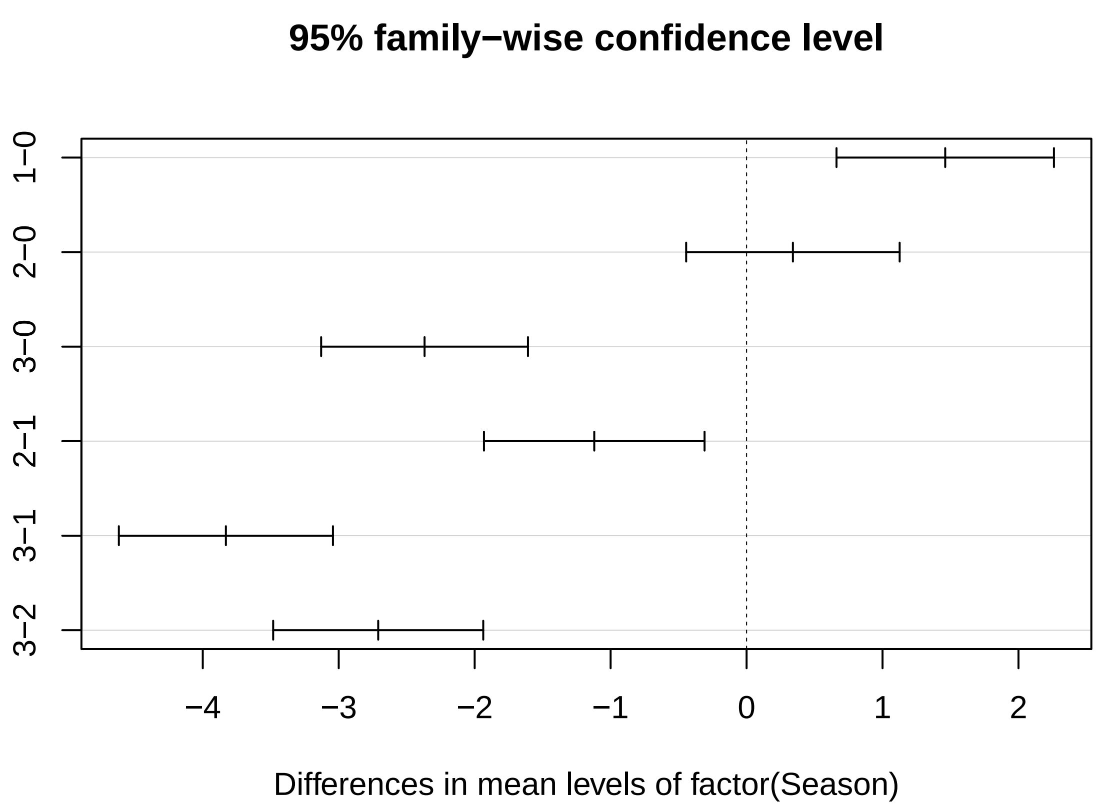
At the 5% significance level there is enough evidence to reject the null hypothesis conclude that a difference exists among the mean mileage per gallon of gas during different seasons. The p-value is a lot less than 0.05 so we reject the null hypothesis of them being the same.

Next lets see which seasons are different.

```r
TukeyHSD(milesAnova)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = MPG ~ factor(Season), data = carData)
##
## $`factor(Season)`
##           diff        lwr        upr     p adj
## 1-0  1.4612198  0.6614433  2.2609962 0.0000183
## 2-0  0.3408412 -0.4443430  1.1260254 0.6785382
## 3-0 -2.3686931 -3.1293204 -1.6080658 0.0000000
## 2-1 -1.1203785 -1.9316127 -0.3091444 0.0022670
## 3-1 -3.8299129 -4.6174029 -3.0424229 0.0000000
## 3-2 -2.7095344 -3.4822001 -1.9368686 0.0000000
```

```r
plot(TukeyHSD(milesAnova))
```



There is enough evidence to show a difference in gas mileage between spring and summer, fall and winter, summer and winter, summer and fall, and spring and winter. There is not enough evidence to show a difference in gas prices between spring and fall since this confidence interval contains zero.

This is the same conclusion we came to about the differences in price per gallon.

Now we will look at the annualized miles based on the car's ownership status. The data has been annualized to show the number of miles driven per year under each status of ownership, and has been estimated based off of total dates since the ownership status of the car has changed mid-year. Note that the ownership status has been recorded as follows:

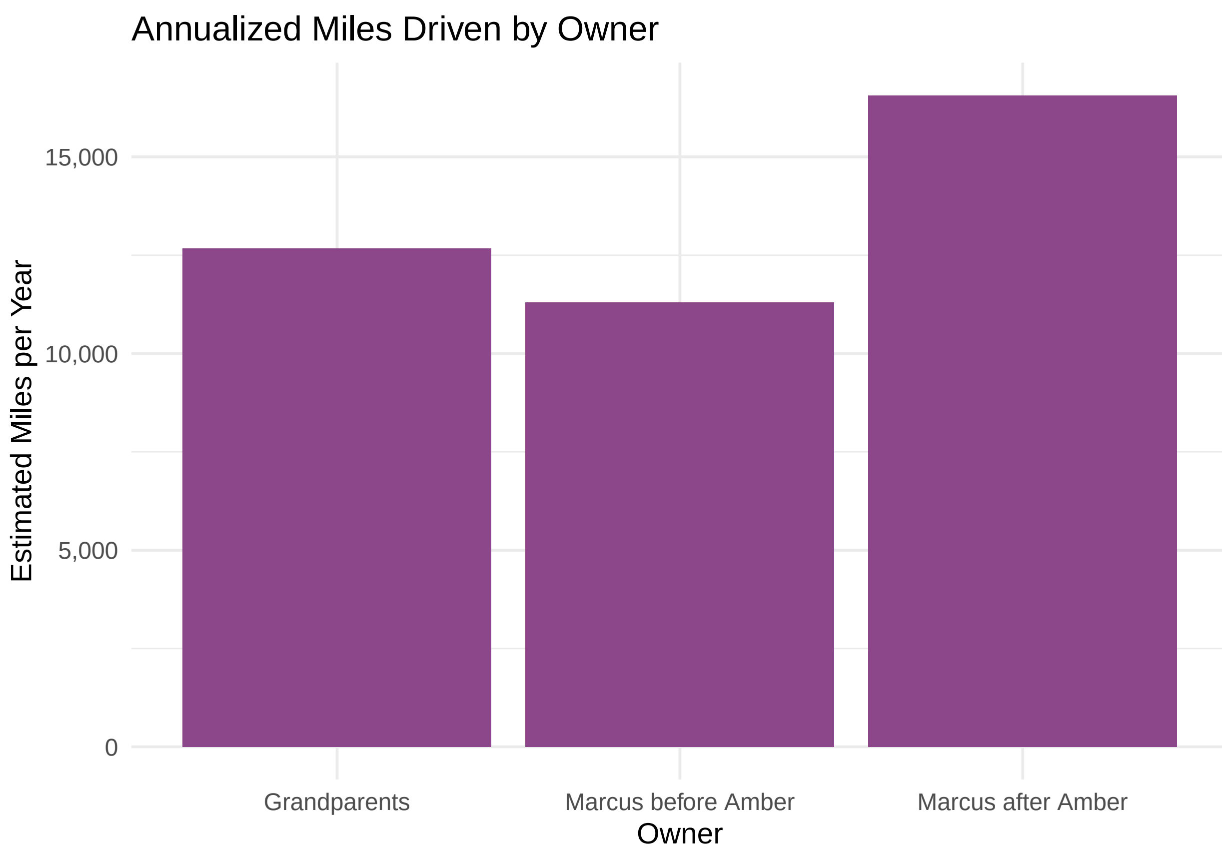
0 = Marcus's grandparents
1 = Marcus before Amber
2 = Marcus after Amber
The ownership status of the vehicle before and after Marcus met Amber also represents when he started regularly driving to Brown Deer and Sturgeon Bay for work.

```
carData$Date <- as.Date(carData$Date)

owner_miles <- carData %>%
  filter(!is.na(Milage)) %>%
  group_by(Owner) %>%
  summarise(
    first_date = min(Date),
    last_date = max(Date),
    total_days = as.numeric(last_date - first_date),
    first_milage = min(Milage),
    last_milage = max(Milage),
    total_miles = as.numeric(last_milage - first_milage),
    miles_per_day = total_miles / total_days,
    annualized_miles = miles_per_day * 365
  )
owner_miles
```

```
## # A tibble: 3 x 9
##   Owner first_date last_date  total_days first_milage last_milage total_miles
##   <int> <date>     <date>          <dbl>        <int>       <int>       <dbl>
## 1     0 2014-03-07 2022-08-12       3080          574      107473      106899
## 2     1 2022-08-24 2024-12-15        844       107782      133926       26144
## 3     2 2024-12-27 2026-04-25        484       134168      156135       21967
## # i 2 more variables: miles_per_day <dbl>, annualized_miles <dbl>
```

```
carData_with_intervals <- carData %>%
  arrange(Owner, Date) %>%
  group_by(Owner) %>%
  mutate(
    days_diff = as.numeric(Date - lag(Date)),
    miles_diff = Milage - lag(Milage),
    miles_per_day = miles_diff / days_diff
  ) %>%
  filter(!is.na(miles_per_day), days_diff > 0) %>%
  ungroup()


ggplot(owner_miles, aes(x = factor(Owner), y = annualized_miles)) +
  geom_col(fill = "orchid4") +
  labs(
    title = "Annualized Miles Driven by Owner",
    x = "Owner",
    y = "Estimated Miles per Year"
  ) +
  scale_x_discrete(
    labels = c(
      "0" = "Grandparents",
      "1" = "Marcus before Amber",
      "2" = "Marcus after Amber"
```

```
    )
  ) +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```



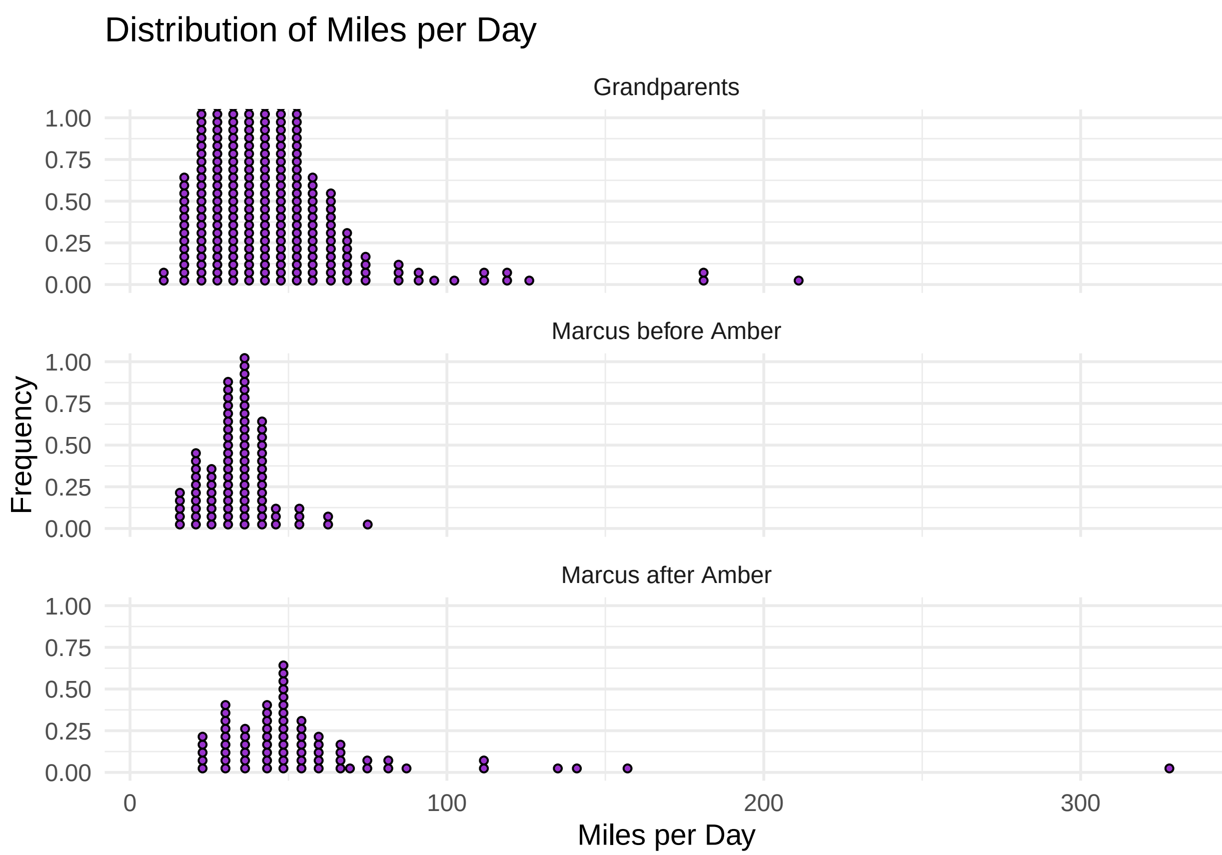
From this bar graph we can see there may be a significant difference in driving distance between who was driving the car. Lets find out by using a Wilcoxon Rank Sum Test. First we need to check our assumptions.

First, we know that the samples are both independent and continuous, meeting the first two assumptions

Next, we will look at the following dotplot to check if the data is not normal or symmetric and if the groups have similar spread.

```
ggplot(carData_with_intervals, aes(x = miles_per_day)) +
  geom_dotplot(
    binaxis = "x",
    fill = "darkorchid",
    color = "black",
    dotsize = 0.5,
    binwidth = 5
  ) +
  facet_wrap(~ Owner, ncol = 1,
             labeller = labeller(Owner = c(
               "0" = "Grandparents",
               "1" = "Marcus before Amber",
               "2" = "Marcus after Amber"
             ))) +
  labs(
```

```r
    title = "Distribution of Miles per Day",
    x = "Miles per Day",
    y = "Frequency"
  ) +
  theme_minimal()
```



From looking at this dot plot we can see that the distribution of miles per day is clearly right-skewed for each group. Because the data is similar between all three and not normal or symmetric, our final assumptions that the data is not normal and have similar spread are satisfied. Therefore, a Wilcoxon Rank Sum test is appropriate.

To assess whether driving behavior differs between ownership groups, we analyze miles driven per day between fill-ups under the following hypotheses:

H0: The distributions of miles driven per day are the same between each two ownership groups Ha: The distributions differ between each two ownership groups

```r
wilcox.test(
  miles_per_day ~ Owner,
  data = filter(carData_with_intervals, Owner %in% c(0, 1))
)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  miles_per_day by Owner
## W = 25235, p-value = 0.03426
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(
  miles_per_day ~ Owner,
  data = subset(carData_with_intervals, Owner %in% c(1, 2))
)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  miles_per_day by Owner
## W = 1322, p-value = 6.7e-10
## alternative hypothesis: true location shift is not equal to 0
```

```r
wilcox.test(
  miles_per_day ~ Owner,
  data = subset(carData_with_intervals, Owner %in% c(0, 2))
)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  miles_per_day by Owner
## W = 10214, p-value = 3.18e-09
## alternative hypothesis: true location shift is not equal to 0
```

At the 5% significance level, there is sufficient evidence in the first test to reject the null hypothesis and conclude that the distribution of miles driven per day differs between the grandparents and Marcus before Amber. But using a 98% significance level, there would not be sufficient evidence to conclude this.

However, there is extremely strong evidence in the second test to reject the null hypothesis and conclude that miles per day increased after Marcus began driving more frequently.

Likewise, there is also extremely strong evidence in the third test to reject the null hypothesis and conclude that Marcus after Amber drives significantly more miles per day than his grandparents.

These conclusions align with the change in usage, that as Marcus began traveling longer distances for work and visiting Manitowoc more frequently.

Conclusion

This study has several limitations that should be considered when interpreting the results. First, the DPG values reflect the actual price paid at the time of purchase and do not account for discounts, rewards programs, or external price adjustments, meaning they may not perfectly represent true market prices. Also, the calculation of MPG assumes that each entry corresponds to a full fill-up. But partial fill-ups may have occurred and could increase or distort MPG.

The data set is also obviously limited to a single vehicle and reflects only the driving habits of Marcus and his grandparents, so the findings cannot be generalized beyond this specific context. Finally, while averages by year and season provide useful summaries, they reduce the granularity of the data and may obscure more detailed patterns that exist at the level of individual trips or shorter time intervals.

Looking forward, there are many opportunities to extend this analysis. Additional research questions could explore more nuanced relationships within the data, such as the interaction between driving patterns and external factors like weather or fuel price volatility. More advanced statistical techniques, including predictive modeling, could be used to estimate future values such as expected MPG, DPG, or total miles driven in upcoming years. These extensions would allow the dataset to move beyond descriptive and inferential analysis toward forecasting and decision-making applications.

Overall, this analysis shows that both fuel efficiency and gas prices vary significantly across seasons, in particular that both gas prices and gas milage are lower in winter than in summer. It also shows that driving behavior has changed substantially over the different periods of the car's ownership. It shows a particularly significant increase in daily driving distance as a result of increased travel for work and personal commitments under Marcus's ownership. These findings show how real world lifestyle changes are reflected clearly in long term observational data.